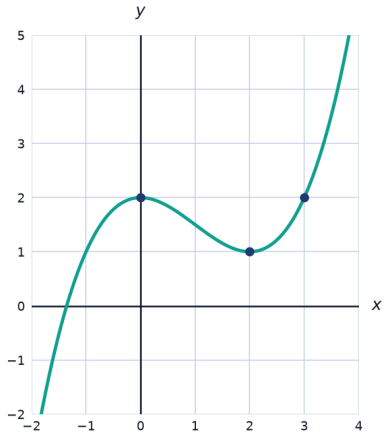
# Mensuration

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Percentage of quantities

For the model $f(x) = 3x + 6$ in percentage of quantities, evaluate $f(5)$.

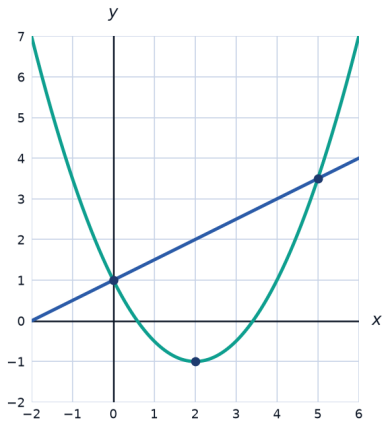



**2** [Foundational] • Target: Percentage of quantities

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

**3** [Foundational] • Target: Percentages with technology

For the model $f(x) = 4x + 7$ in percentages with technology, evaluate $f(2)$.




**4** [Foundational] • Target: Percentages with technology

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

**5** [Foundational] • Target: Percentages with technology

A data set recorded for percentages with technology consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

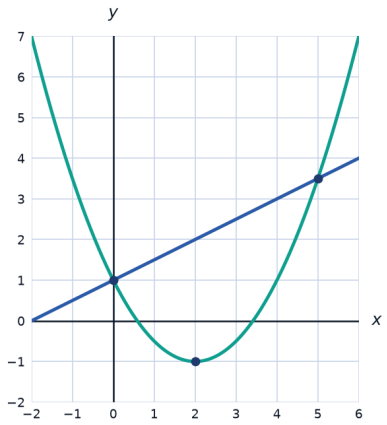
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**6** [Foundational] • Target: Percentages with technology

For the model $f(x) = 4x + 7$ in percentages with technology, evaluate $f(3)$.

**7** [Foundational] • Target: Percentages with technology

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.




## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Percentages with technology

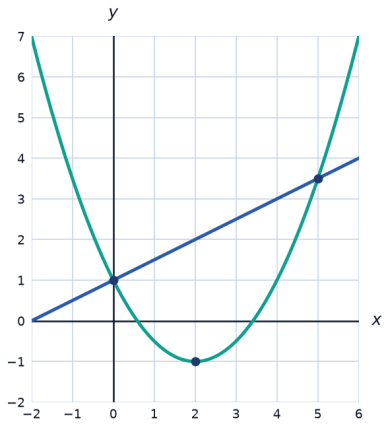
For the model $f(x) = 4x + 7$ in percentages with technology, evaluate $f(4)$.

**9** [Intermediate] • Target: Percentages with technology

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

**10** [Intermediate] • Target: Percentages with technology

For the model $f(x) = 4x + 7$ in percentages with technology, evaluate $f(5)$.



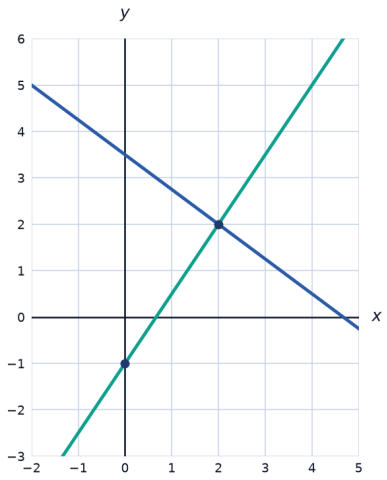

**11** [Intermediate] • Target: Percentages with technology

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

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**12** [Intermediate] • Target: Finding the whole (Extended)

For the model $f(x) = 5x + 8$ in finding the whole (extended), evaluate $f(2)$.




**13** [Intermediate] • Target: Finding the whole (Extended)

A data set recorded for finding the whole (extended) consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

**14** [Intermediate] • Target: Finding the whole (Extended)

For the model $f(x) = 5x + 8$ in finding the whole (extended), evaluate $f(3)$.

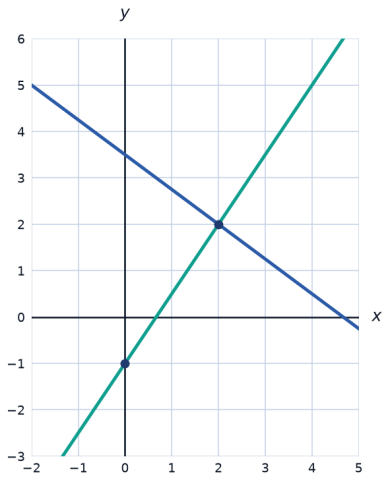
## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Finding the whole (Extended) • Cross-Lesson: 1.01 Number

For the model $f(x) = 5x + 8$ in finding the whole (extended), evaluate $f(4)$.

**16** [Challenging] • Target: Finding the whole (Extended) • Cross-Lesson: 10.01 Sets & Venn Diagrams

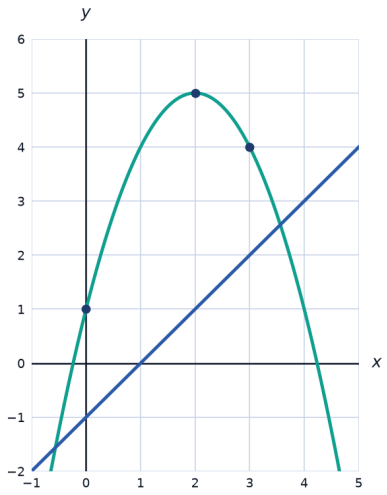
For the model $f(x) = 5x + 8$ in finding the whole (extended), evaluate $f(5)$.




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**17** [Challenging] • Target: Percentage increase and decrease • Cross-Lesson: 2.01 Algebra & Graphs

For the model $f(x) = 6x + 9$ in percentage increase and decrease, evaluate $f(2)$.




**18** [Challenging] • Target: Percentage increase and decrease • Cross-Lesson: 3.01 Coordinate Geometry

A data set recorded for percentage increase and decrease consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

**19** [Challenging] • Target: Percentage increase and decrease • Cross-Lesson: 5.01 Geometry

For the model $f(x) = 6x + 9$ in percentage increase and decrease, evaluate $f(3)$.

**20** [Challenging] • Target: Percentage increase and decrease • Cross-Lesson: 6.01 Trigonometry

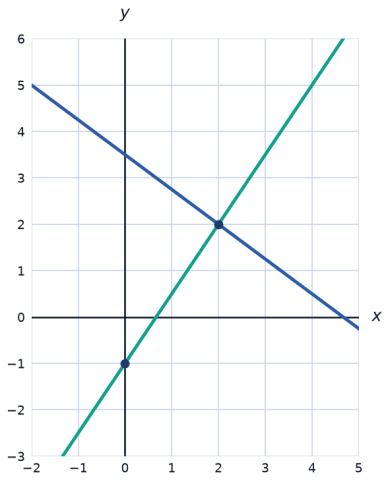
For the model $f(x) = 6x + 9$ in percentage increase and decrease, evaluate $f(4)$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: Converting between percentages, fractions, and decimals

For the model $f(x) = 6x + 4$ in converting between percentages, fractions, and decimals, evaluate $f(2)$.




**a.** 16

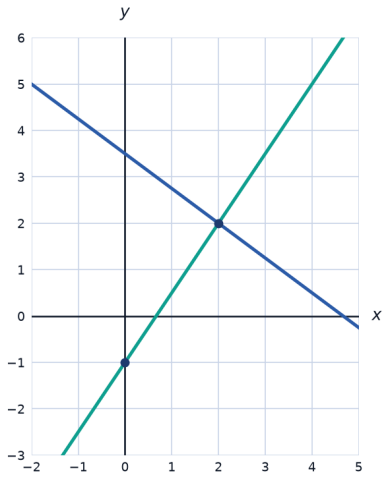
**b.** 22

**c.** 12

**d.** 20

**22** [Foundational] • Target: Converting between percentages, fractions, and decimals

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.




**a.** $x = 2$ and $x = 6$

**b.** $x = 2$ and $x = 5$

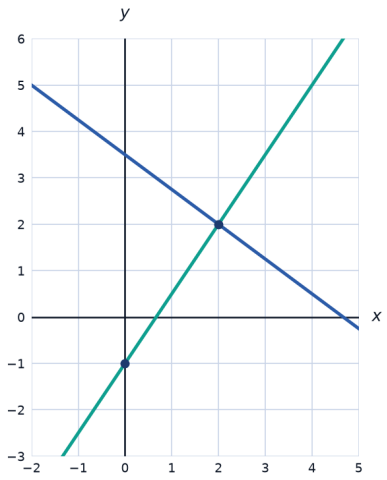
**c.** $x = -2$ and $x = -5$

**d.** $x = 7$ and $x = 10$

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**23** [Foundational] • Target: Converting between percentages, fractions, and decimals

For the model $f(x) = 6x + 4$ in converting between percentages, fractions, and decimals, evaluate $f(5)$.




**a.** 34

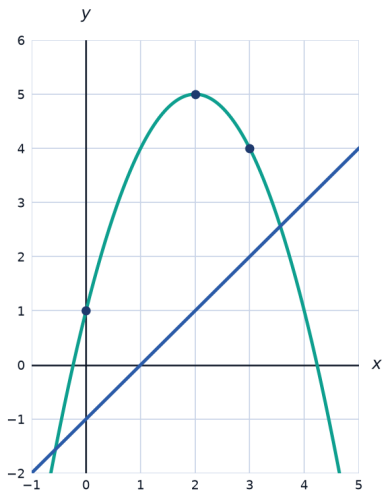
**b.** 40

**c.** 30

**d.** 38

**24** [Foundational] • Target: Quantities as percentages

For the model $f(x) = 2x + 5$ in quantities as percentages, evaluate $f(2)$.




**a.** 9

**b.** 11

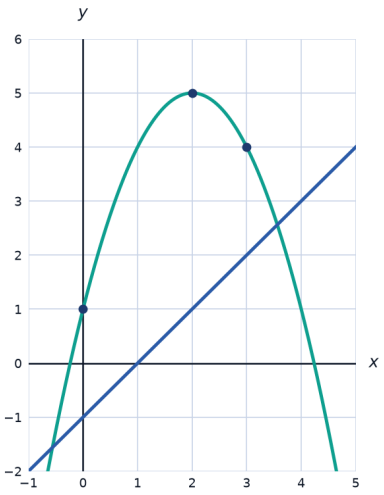
**c.** 4

**d.** 13

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**25** [Foundational] • Target: Quantities as percentages

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.




**a.** $x = 2$ and $x = 7$

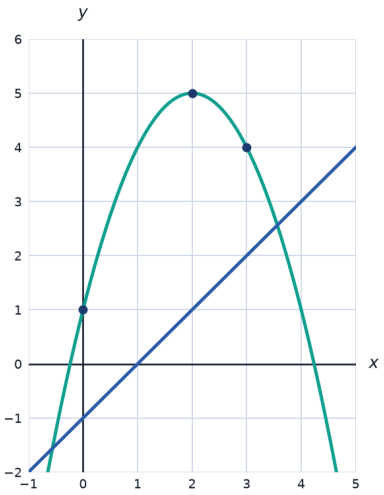
**b.** $x = 2$ and $x = 6$

**c.** $x = -2$ and $x = -6$

**d.** $x = 8$ and $x = 12$

**26** [Foundational] • Target: Quantities as percentages

For the model $f(x) = 2x + 5$ in quantities as percentages, evaluate $f(5)$.




**a.** 15

**b.** 17

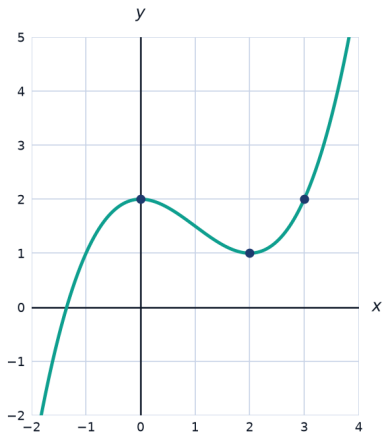
**c.** 10

**d.** 19

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**27** [Foundational] • Target: Percentage of quantities

For the model $f(x) = 3x + 6$ in percentage of quantities, evaluate $f(2)$.




**a.** 12

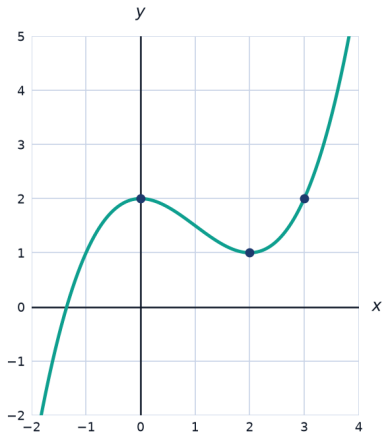
**b.** 15

**c.** 6

**d.** 16

**28** [Foundational] • Target: Percentage of quantities

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.




**a.** $x = 2$ and $x = 8$

**b.** $x = 2$ and $x = 7$

**c.** $x = -2$ and $x = -7$

**d.** $x = 9$ and $x = 14$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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**29** [Intermediate] • Target: Converting between percentages, fractions, and decimals

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

**a.** $x = 1$ and $x = 5$

**b.** $x = 1$ and $x = 4$

**c.** $x = -1$ and $x = -4$

**d.** $x = 5$ and $x = 4$

**30** [Intermediate] • Target: Converting between percentages, fractions, and decimals

A data set recorded for converting between percentages, fractions, and decimals consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

**a.** 16.5

**b.** 17.5

**c.** 18.5

**d.** 20.5

**31** [Intermediate] • Target: Converting between percentages, fractions, and decimals

For the model $f(x) = 6x + 4$ in converting between percentages, fractions, and decimals, evaluate $f(3)$.

**a.** 22

**b.** 28

**c.** 18

**d.** 26

**32** [Intermediate] • Target: Converting between percentages, fractions, and decimals

For the model $f(x) = 6x + 4$ in converting between percentages, fractions, and decimals, evaluate $f(4)$.

**a.** 28

**b.** 34

**c.** 24

**d.** 32

**33** [Intermediate] • Target: Converting between percentages, fractions, and decimals

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

**a.** $x = 3$ and $x = 7$

**b.** $x = 3$ and $x = 6$

**c.** $x = -3$ and $x = -6$

**d.** $x = 9$ and $x = 18$

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**34** [Intermediate] • Target: Converting between percentages, fractions, and decimals

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

**a.** $x = 4$ and $x = 8$

**b.** $x = 4$ and $x = 7$

**c.** $x = -4$ and $x = -7$

**d.** $x = 11$ and $x = 28$

**35** [Intermediate] • Target: Quantities as percentages

A data set recorded for quantities as percentages consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

**a.** 17.5

**b.** 18.5

**c.** 19.5

**d.** 21.5

**36** [Intermediate] • Target: Quantities as percentages

For the model $f(x) = 2x + 5$ in quantities as percentages, evaluate $f(3)$.

**a.** 11

**b.** 13

**c.** 6

**d.** 15

**37** [Intermediate] • Target: Quantities as percentages

For the model $f(x) = 2x + 5$ in quantities as percentages, evaluate $f(4)$.

**a.** 13

**b.** 15

**c.** 8

**d.** 17

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Quantities as percentages • Cross-Lesson: 3.01 Coordinate Geometry

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

**a.** $x = 3$ and $x = 8$

**b.** $x = 3$ and $x = 7$

**c.** $x = -3$ and $x = -7$

**d.** $x = 10$ and $x = 21$

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**39** [Challenging] • Target: Quantities as percentages • Cross-Lesson: 5.01 Geometry

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

**a.** $x = 4$ and $x = 9$

**b.** $x = 4$ and $x = 8$

**c.** $x = -4$ and $x = -8$

**d.** $x = 12$ and $x = 32$

**40** [Challenging] • Target: Percentage of quantities • Cross-Lesson: 6.01 Trigonometry

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

**a.** $x = 1$ and $x = 7$

**b.** $x = 1$ and $x = 6$

**c.** $x = -1$ and $x = -6$

**d.** $x = 7$ and $x = 6$

**41** [Challenging] • Target: Percentage of quantities • Cross-Lesson: 7.01 Transformations & Vectors

For the model $f(x) = 3x + 6$ in percentage of quantities, evaluate $f(3)$.

**a.** 15

**b.** 18

**c.** 9

**d.** 19

**42** [Challenging] • Target: Percentage of quantities • Cross-Lesson: 8.01 Probability & Statistics

For the model $f(x) = 3x + 6$ in percentage of quantities, evaluate $f(4)$.

**a.** 18

**b.** 21

**c.** 12

**d.** 22

**43** [Challenging] • Target: Percentage of quantities • Cross-Lesson: 9.01 Functions

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

**a.** $x = 3$ and $x = 9$

**b.** $x = 3$ and $x = 8$

**c.** $x = -3$ and $x = -8$

**d.** $x = 11$ and $x = 24$

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**44** [Challenging] • Target: Percentages with technology • Cross-Lesson: 1.01 Number

A data set recorded for percentages with technology consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

**a.** 19.5

**b.** 20.5

**c.** 21.5

**d.** 23.5

**45** [Challenging] • Target: Finding the whole (Extended) • Cross-Lesson: 10.01 Sets & Venn Diagrams

A data set recorded for finding the whole (extended) consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

**a.** 20.5

**b.** 21.5

**c.** 22.5

**d.** 24.5

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